Koneru Lakshmaiah Education Foundation
(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)
Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.
Phone No: 7815926816, www.klh.edu.in

DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: SOA PROGRAMMING AND MICROSERVICES– (24SDCS03)
A.Y. 2026 - 2027

TITLE OF THE PROJECT	AegisCore: An Intelligent API Security Monitoring & Threat Intelligence Platform	
1	2420030113	Harshini
2	2420030373	Pallavi
3	2420030481	Divija
Name of the Guide		**Dr.Venkateswari Ketineni**

Abstract:

Application Programming Interfaces (APIs) are a critical part of modern software systems, connecting web applications, mobile applications, microservices, cloud services, and external platforms. As API usage increases, security threats such as unauthorized access, credential abuse, injection attacks, abnormal request patterns, API abuse, and data exfiltration have become important challenges. Traditional monitoring approaches often depend on static rules and manual log analysis, which can make it difficult to identify evolving and previously unseen threats in real time. This highlights the need for an intelligent platform that continuously monitors API activity and assists security teams in detecting and investigating suspicious behavior.

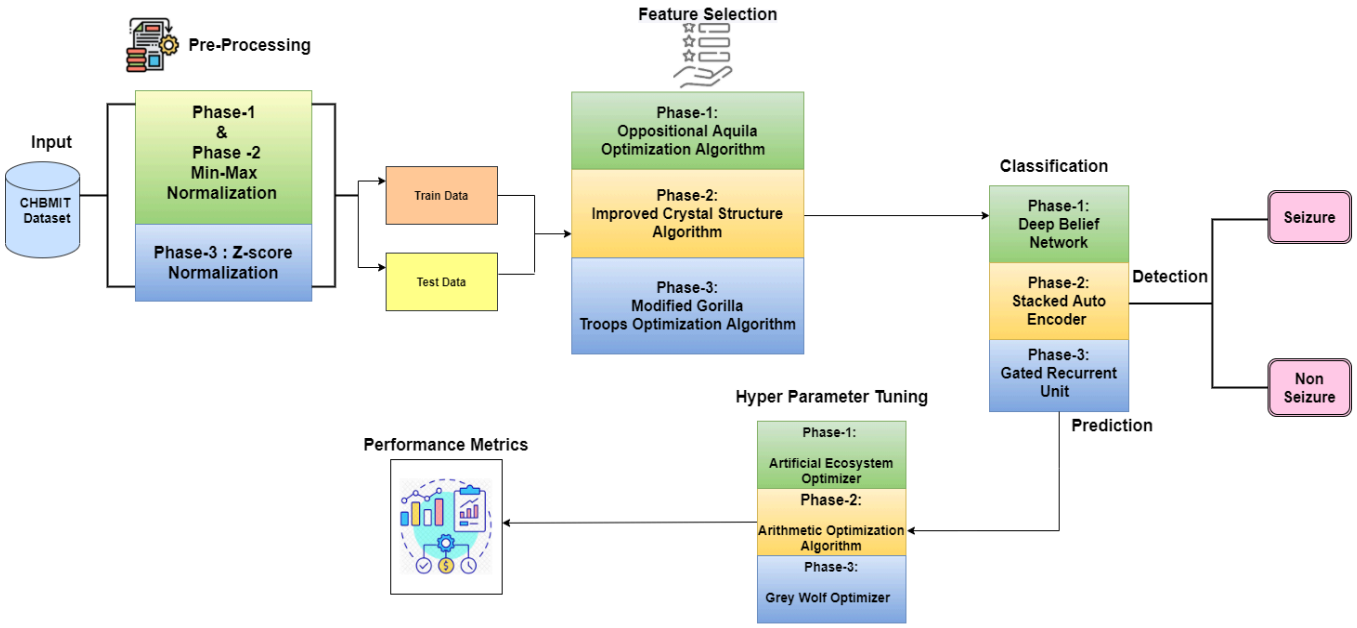
AegisCore is proposed as an intelligent API security monitoring and threat intelligence platform designed to provide continuous visibility into API traffic and security events. The system collects API gateway and application logs, performs preprocessing and normalization, and extracts security-relevant features such as request frequency, response

status patterns, authentication failures, source information, endpoint behavior, and unusual access sequences. A machine-learning-based anomaly detection component analyzes these features to identify deviations from normal API behavior and classify potentially malicious activity. The platform also maintains a threat intelligence layer that correlates suspicious indicators, IP addresses, request patterns, and known threat information to improve the context of detected incidents.

The proposed platform follows a modular architecture consisting of API traffic collection, log processing, feature extraction, anomaly and threat detection, threat intelligence correlation, alert generation, and security visualization. A monitoring dashboard presents API health, traffic trends, detected anomalies, risk levels, frequently targeted endpoints, and security alerts to support faster investigation and response. The system can additionally provide risk scoring and incident summaries so that security teams can prioritize high-impact events. By combining continuous API monitoring, intelligent anomaly detection, and threat intelligence, AegisCore aims to improve the visibility, accuracy, and responsiveness of API security operations while supporting a scalable microservices-oriented environment.

Project Architecture/Project Overview:

Project Overview: API requests are collected from API gateways and services, then processed and normalized for analysis. Relevant behavioral and security features are extracted and passed to anomaly-detection models. Detected events are correlated with threat intelligence to generate risk scores and alerts. The resulting incidents and API security metrics are presented through a monitoring dashboard for investigation and response.


Input
CHBMIT Dataset
Pre-Processing
Phase-1 & Phase -2 Min-Max Normalization
Phase-3 : Z-score Normalization
Train Data
Test Data
Feature Selection
Phase-1: Oppositional Aquila Optimization Algorithm
Phase-2: Improved Crystal Structure Algorithm
Phase-3: Modified Gorilla Troops Optimization Algorithm
Classification
Phase-1: Deep Belief Network
Phase-2: Stacked Auto Encoder
Phase-3: Gated Recurrent Unit
Detection
Seizure
Non Seizure
Prediction
Hyper Parameter Tuning
Phase-1: Artificial Ecosystem Optimizer
Phase-2: Arithmetic Optimization Algorithm
Phase-3: Grey Wolf Optimizer
Performance Metrics